

Refrigerated air dryers

WRD series

Compressed air leaves the compressor saturated with water. WRD dryers chill it to a +3°C pressure dew point so the moisture condenses and drains out before it reaches your machines, valves and finished product.



Capacity	Pressure dew point	Inlet air pressure	Inlet air temperature
20 to 2,000 CFM listed	+3 °C	3 to 16 bar g	10 to 60 °C

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Specifications online

Layout edition 2026-10 · Product-range information. Confirm the selected model and duty with Win Equipments before ordering.

Models & rating conditions

Rating conditions

Flow rating	ISO 7183, FAD at 20 °C, 1 bar	Max. working pressure	16 bar g
Pressure drop	≤ 0.2 bar g	Supply	180–260 V 1 Ph · 380–420 V 3 Ph

Capacity & operating data

Model	Capacity CFM	Capacity m³/hr	Refrigeration compressor kW	Electrical V / Ph / Hz
WRD 20 S	20	34	0.13	230 / 1 / 50
WRD 40 S	40	68	0.21	230 / 1 / 50
WRD 50 S	50	85	0.28	230 / 1 / 50
WRD 60 S	60	102	0.32	230 / 1 / 50
WRD 80 S	80	136	0.40	230 / 1 / 50
WRD 100 S	100	170	0.67	230 / 1 / 50
WRD 150 S	150	255	0.85	230 / 1 / 50
WRD 200 S	200	340	1.2	230 / 1 / 50
WRD 250 S	250	425	1.35	230 / 1 / 50
WRD 300 T	300	510	1.7	440 / 3 / 50
WRD 400 T	400	680	2.0	440 / 3 / 50
WRD 500 T	500	850	2.3	440 / 3 / 50
WRD 600 T	600	1020	2.72	440 / 3 / 50
WRD 750 T	750	1275	3.3	440 / 3 / 50
WRD 1000 T	1000	1700	4.5	440 / 3 / 50
WRD 1500 T	1500	2550	6.2	440 / 3 / 50
WRD 2000 T	2000	3400	7.8	440 / 3 / 50

Dimensions & connections

Rating conditions

Flow rating	ISO 7183, FAD at 20 °C, 1 bar	Max. working pressure	16 bar g
Pressure drop	≤ 0.2 bar g	Supply	180–260 V 1 Ph · 380–420 V 3 Ph

Construction, dimensions & connections

Model	Dimensions L × W × H mm	Condenser	End connections BSP / NB
WRD 20 S	550 × 400 × 600	Air	1/2"
WRD 40 S	550 × 400 × 600	Air	1/2"
WRD 50 S	540 × 500 × 600	Air	3/4"
WRD 60 S	540 × 500 × 800	Air	3/4"
WRD 80 S	540 × 500 × 800	Air	1"
WRD 100 S	700 × 500 × 800	Air	1"
WRD 150 S	700 × 500 × 1000	Air	1 1/2"
WRD 200 S	700 × 750 × 1000	Air	1 1/2"
WRD 250 S	900 × 750 × 1000	Air	2"
WRD 300 T	900 × 750 × 1200	Air	2"
WRD 400 T	950 × 1700 × 1200	Air	2 1/2"
WRD 500 T	1450 × 1700 × 1250	Air	2 1/2"
WRD 600 T	1450 × 1700 × 2100	Air / Water	2 1/2"
WRD 750 T	1450 × 1700 × 2100	Air / Water	3" NB flange
WRD 1000 T	1450 × 1700 × 2100	Air / Water	4" NB flange
WRD 1500 T	1450 × 1700 × 2100	Air / Water	5" NB flange
WRD 2000 T	1450 × 1700 × 2100	Air / Water	6" NB flange

- Model code: WRD + capacity in CFM + S (single phase) or T (three phase). Working pressure 16 bar g max.
- Flow capacities per ISO 7183: air suction of FAD at 20 °C, 1 bar.
- Rated power is the maximum power consumed at rated conditions.
- Voltage range 180–260 V for 1 Ph and 380–420 V for 3 Ph.

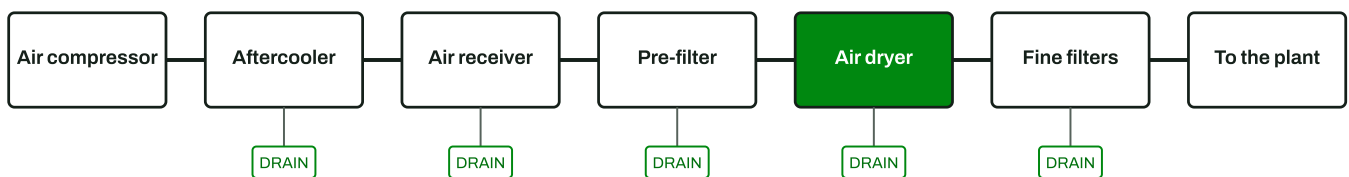
Working principle & installation

How it works

Air contains dust, moisture and traces of oil. Untreated wet compressed air causes equipment failures and contaminates the production process, so it must be dried before use.

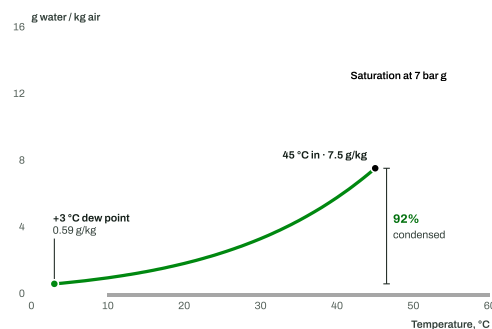
- 1 Pre-cool**
Saturated compressed air enters the pre-cooler / re-heater, where the outgoing cold air pre-cools it and reduces the heat load on the evaporator.
- 2 Chill**
In the evaporator the air is cooled further to the required dew point. The refrigerant compressor and condenser supply low-temperature refrigerant to the evaporator.
- 3 Separate**
Cooling reduces the air's capacity to hold water vapour; the moisture condenses and is removed by the water separator and automatic drain.
- 4 Re-heat**
The cold, dry air is re-heated as it passes back through the air-to-air heat exchanger before leaving the dryer.
- 5 Match load**
The thermostatic expansion valve and hot gas bypass valve match the refrigerant system to the compressed air cooling load, keeping the dew point constant.

Typical compressed air installation



Typical arrangement only. Confirm filtration sequence and condensate handling for your application.

Water in compressed air at 7 bar g



Saturated air at 45 °C holds about 7.5 g of water per kg; at +3 °C pressure dew point, about 0.6 g/kg. About 92% must condense and drain. Calculated values.

Construction & selection questions

Construction & features

Co-axial heat exchanger

- Acts as pre-cooler and evaporator
- Copper tubes in a coiled tube-in-tube arrangement
- Fully encapsulated in PUF insulation

Thermostatic expansion valve

- Selected for the cooling load and operating condition
- Holds a constant dew point at varying load
- Responds to suction pressure and temperature

Hot gas bypass valve

- Fitted between compressor discharge and evaporator
- Prevents evaporator temperature dropping below 5 °C
- Manual or automatic

Refrigeration compressor

- Hermetically sealed reciprocating or scroll compressor
- Suitable for eco-friendly gases
- Low noise and low power consumption

Temperature controller

- Programmed micro-controller
- Indicates dew point, inlet and condensing temperature with alarms
- On-delay timer protects the compressor

Condenser

- Air-cooled: aluminium plate-finned cross-flow exchanger
- Water-cooled option on larger models
- Spigot construction lowers pressure drop and compressor power

Applications: Automobile · Cement · Chemical · CNC machine shops · Electronics · Food processing · Foundry · Glass · Leather · Textile · Paper mills · Rice mills · Power plants · Printing

What dew point does a WRD dryer deliver?

WRD dryers deliver a +3 °C pressure dew point and hold it at varying load using a thermostatic expansion valve and hot gas bypass valve.

Which models run on single-phase power?

Models with an S suffix (WRD 20 S to WRD 250 S) run on 230 V single phase. T models from WRD 300 T upward run on three-phase supply.

How do I size a refrigerated dryer for my compressor?

Divide your compressor's actual capacity by the product of the four correction factors for inlet temperature, ambient temperature, inlet pressure and dew point. Then choose the WRD model at or above that nominal capacity. Use the sizing calculator on winequipments.com.

Is a water-cooled condenser available?

Yes. WRD 600 T and larger are offered with air- or water-cooled condensers.

Selection & applications

Sizing for your conditions

$$\text{Dryer nominal capacity} = \text{Compressor actual capacity} \div (C1 \times C2 \times C3 \times C4)$$

C1 · Inlet temperature

°C	Factor
30	1.2
35	1.15
40	1.05
45	1.0
50	0.85
55	0.8
60	0.7

C2 · Ambient temperature

°C	Factor
25	1.2
30	1.15
35	1.0
40	0.91
45	0.87
50	0.78

C3 · Inlet pressure

bar g	Factor
4	0.75
5	0.85
6	0.93
7	1.0
8	1.06
9	1.11
10.5	1.15
11	1.18
12	1.2
13	1.22
14	1.23
15	1.25

C4 · Dew point

°C	Factor
3	1.0
5	1.09
7	1.15
8	1.18
10	1.3

Confirm your selection

Send the model, required capacity, operating temperatures, pressure and site conditions. For a model-specific datasheet or GA drawing, include your installation requirement.

Win Equipments

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Catalogue values are for product selection. Confirm duty, materials, connections and final dimensions with Win Equipments before ordering. Layout edition 2026-10.